

A Monolithically Integrated 100-GHz Optical Receiver Demonstrating 224-GBd PAM-4 Operation

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Abstract We demonstrate a C-band optical receiver monolithically integrating a differential TIA and photodiodes in a 250-nm SiGe BiCMOS EPIC platform. The receiver achieves a bandwidth beyond 100 GHz and optical-amplifier-free 224-GBd PAM-4 operation using only linear equalization, supporting net 400-Gb/s-per-lane transmission while consuming 162 mW. ©2026 The Author(s)

Introduction

Driven by the rapid growth of AI workloads and the associated expansion of datacenter infrastructure, short-reach optical interconnect rates are being pushed to 400 Gb/s per lane. Several components targeting 200-GBd PAM-4 links have been showcased already, including modulators [1, 2] and photodiodes [3–5] demonstrating analog bandwidths beyond 100 GHz, as well as the required SerDes ICs [6, 7].

At the receiver side, the photodiode (PD) and transimpedance amplifier (TIA) typically reside on different chips, with the PD implemented on a dedicated photonic IC (PIC) platform such as InP or silicon photonics, while the TIA is implemented in a dedicated electronic IC (EIC) technology such as CMOS, SiGe BiCMOS or InP. Such a heterogeneous receiver assembly has the advantage that the EIC platform can be fully optimized for transistor performance, while the PIC is fully tailored to photonics. However, at these increasingly high speeds, performance is highly dependent on the parasitics of the PD and the interconnect between the PD and the TIA. As shown in Fig. 1, such heterogeneous implementations inherently introduce parasitic capacitance due to bond pads and TIA input ESD protection, as well as parasitic inductance associated with the chip-to-chip interconnect, regardless of whether wire bonding, flip-chip bonding, hybrid bonding, or transfer printing is used.

In a so-called monolithic process, photonic and electronic capabilities are combined into a single EPIC platform (Fig. 1). This enables direct integration of the PD with the TIA, eliminating the need for bond pads and ESD protection circuitry at that interface. Additionally, peaking inductance can be introduced between the PD and TIA input in a controlled and accurately modeled manner.

Historically, monolithic optical receivers have struggled to support symbol rates of 100 GBd and beyond, both for SOI-CMOS based platforms [8, 9], as well as for platforms based on SiGe BiC-

MOS [10–13]. More recently, monolithic integration of high-speed Ge-fin PDs on a 250-nm BiCMOS platform was demonstrated with photodiode responsivity of 0.45 A/W at bandwidths beyond 110 GHz and SiGe HBTs with $f_T/f_{max} = 390/440$ GHz [14].

Building upon this recently introduced monolithic EPIC platform, this work demonstrates a C-band optical receiver with a bandwidth beyond 100 GHz and a power consumption of only 162 mW, capable of supporting net 400-Gb/s-per-lane operation without optical amplification. To the authors' knowledge, this is the first monolithically integrated optical receiver to experimentally demonstrate 224-GBd PAM-4 operation, establishing the viability of monolithic EPIC platforms for next-generation 400-Gb/s-per-lane interconnects.

Monolithic receiver

The circuit diagram of this chip is depicted in Fig. 2, showing a fully differential TIA with a C-band PD connected to each input. The optical input is coupled into the chip with two grating couplers after which it is guided to the PDs using optical waveguides. The PD cathodes are locally decoupled and biased with an externally applied bias voltage of 3.8 V, resulting in a reverse bias voltage of 2 V across each photodiode. The anodes are connected to the TIA input through 120-pH series peaking inductors. The tap points of these induc-

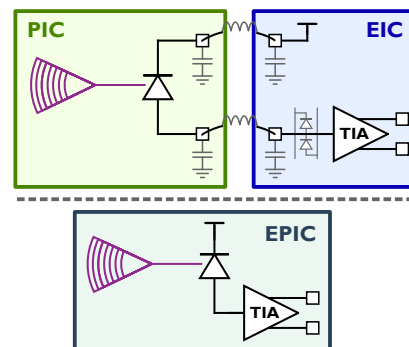


Fig. 1: Comparison between a traditional optical receiver assembly and a monolithically integrated solution.

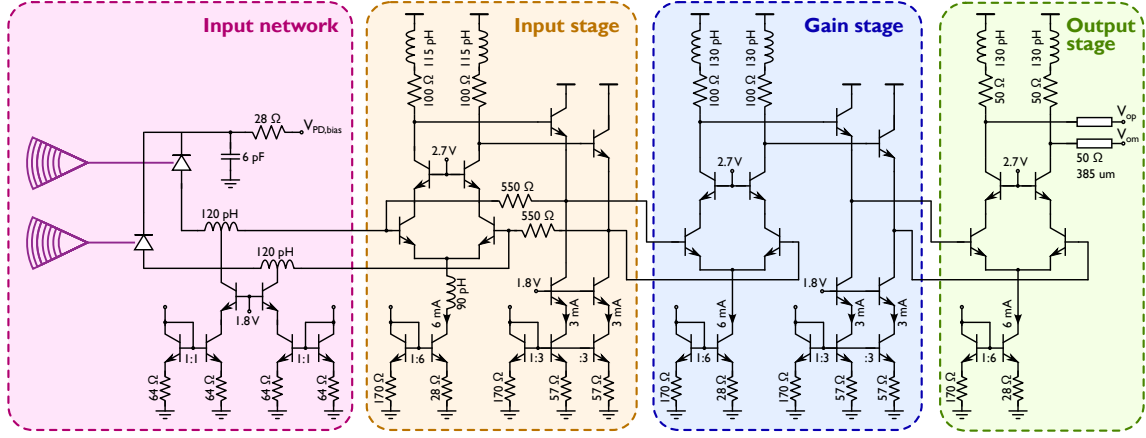


Fig. 2: Circuit diagram of the monolithically integrated differential receiver.

tors enable independent cancellation of the DC photocurrent in each branch using bipolar current mirrors.

The input stage employs a shunt-feedback (SFB) topology in which a cascoded differential pair and subsequent emitter followers form the forward amplifier, using 115-pH shunt peaking inductors to extend its bandwidth and a tail inductor of 90 pH to improve its high-frequency common-mode rejection. Achieving both high bandwidth and low noise requires joint optimization of the forward amplifier sizing, input inductance, and feedback resistance together with the PD parasitics.

The gain stage provides additional amplification while presenting only a minimal load to the output node of the input stage. The output stage mainly serves as an output buffer and provides a 50-Ω output impedance at each output terminal. On-chip transmission lines are used to connect this output to the output bond pads. These final stages consist of cascoded differential pairs with shunt peaking inductors as well, with emitter followers in between.

The entire circuit occupies an area of 0.68 mm² (excluding the grating couplers, which are located elsewhere on the chip), or 0.18 mm² when excluding the output transmission lines and the bond pads. The receiver operates from a single 3 V supply and consumes 54 mA (162 mW). At a net data rate of 400 Gb/s, this corresponds to an energy efficiency of 0.4 pJ/bit.

Fig. 3 shows the simulated post-layout frequency response (which includes the PD model) at the output of each stage. The simulated overall transimpedance gain is 61.8 dBΩ, with a 3-dB bandwidth of 106 GHz and a group-delay variation (GDV) of 11.2 ps. The simulated output return loss remains below -10 dB up to 65 GHz, and increases to -7.5 dB at 100 GHz.

Transmission experiments

To evaluate the performance of this chip as a 400-Gb/s-per-lane direct-detection receiver, it was integrated in the experimental setup of Fig. 4(a). At

the transmit side, a PRBS15-based PAM-4 signal is pulse-shaped by an RRC filter ($\alpha = 0.03$) and upsampled to 256 GSa/s after which a FIR filter is applied for pre-emphasis. A 256 GSa/s arbitrary waveform generator (AWG; Keysight M8199B) drives a 110-GHz thin-film lithium niobate (TFLN) modulator (biased at its quadrature point) which modulates this signal onto a 1550 nm optical carrier. As shown in Fig. 4(b), a fiber probe couples the optical signal into one of the PDs of the monolithic receiver (0 dBm average in-waveguide power), while the other PD remains unilluminated. The receiver output is captured using a 110 GHz differential RF probe and connected directly to the 130-GHz remote sampling heads of a Keysight DCA-X sampling oscilloscope through DC blocks. Finally, offline digital signal processing (DSP) is applied, including timing recovery, resampling to two samples per symbol, matched RRC filtering and linear feed-forward equalization (FFE). No DFE or MLSE is employed. Alternatively, the transmitter output can be applied to a reference PD as well, consisting of a packaged 100 GHz PD (Fraunhofer HHI). No optical amplification was employed in either link.

The resulting PAM-4 eye diagrams for 200, 212.5 and 224 GBd are shown in Fig. 5, along with the histograms at the optimal sampling instant and the counted bit error rate (BER). At 200 GBd (400 Gb/s gross), a BER of 1.4E-4 was recorded, below the threshold for KP4 forward

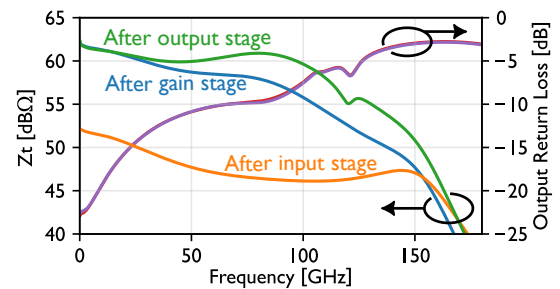


Fig. 3: Simulated transimpedance gain after the input, gain and output stage (left axis) and output return loss (right axis).

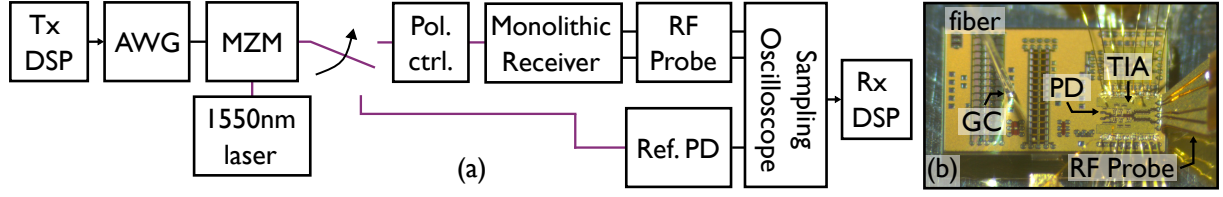


Fig. 4: (a) Experimental setup for the transmission experiments and (b) micrograph of the monolithic receiver.

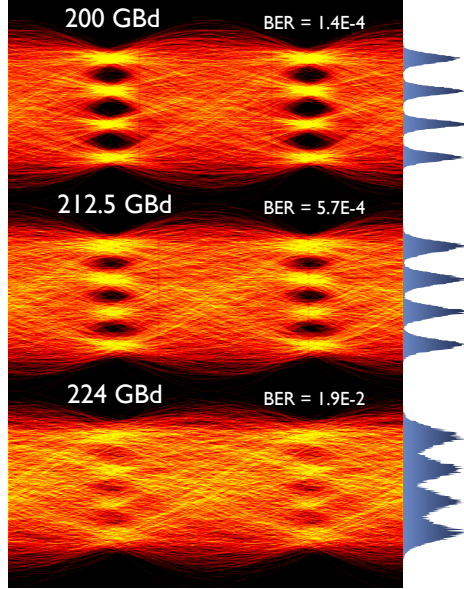


Fig. 5: Received PAM-4 eye diagrams at 200, 212.5 and 224 GBd, with the associated histograms at the optimal sampling instant.

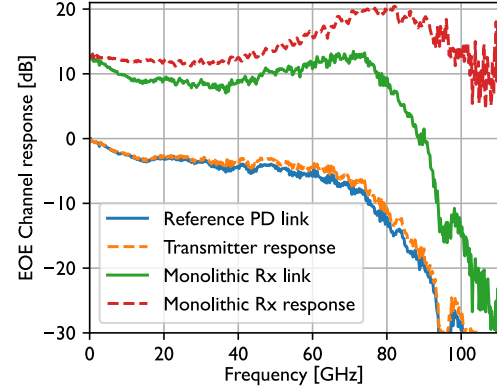


Fig. 6: Frequency response of the experimental link with the monolithic receiver and reference link, and frequency response of the transmitter and the monolithic receiver that were derived from it.

error correction (FEC). At 212.5 GBd, the BER increases slightly to 5.7E-4, remaining below the 6.25% overhead FEC limit and therefore supporting net 400 Gb/s operation. Further increasing the symbol rate to 224 GBd yields a further increased BER of 1.9E-2, which is just below the threshold for 12% overhead FEC, again achieving net 400 Gb/s operation.

Fig. 6 shows the electro-optical-electrical (EOE) frequency response of the entire link and its components, both when using the monolithic receiver as well as with the reference PD. De-embedding the datasheet response of the reference photodiode from the measured reference link response yields the frequency response of the transmitter, which is dominated by the severe rolloff of the AWG beyond 80 GHz. The response of the monolithic receiver itself is subsequently obtained by de-embedding the transmitter response from the measured EOE response of the monolithic receiver link, revealing a 3-dB bandwidth slightly exceeding 100 GHz, in good agreement with the simulations.

The receiver noise was characterized in the absence of an optical input signal, yielding a differential output noise voltage of 9.2 mVrms. Subtracting the noise contribution of the oscilloscope and dividing by the simulated transimpedance gain, an input-referred noise current of 7.4 μ Arms is obtained. Using the simulated 106-GHz bandwidth

as an estimate of the noise bandwidth results in an input-referred noise density of 22.7 pA/ $\sqrt{\text{Hz}}$, corresponding to a noise-equivalent power (NEP) of 50.3 pW/ $\sqrt{\text{Hz}}$ at the PD.

Conclusions

For the first time, a monolithically-integrated optical receiver supporting net 400-Gb/s-per-lane PAM-4 was demonstrated. Symbol rates up to 224 GBd were reached using only linear equalization, with the measured BERs at 212.5 and 224 GBd remaining below the respective FEC thresholds.

The key performance metrics of this receiver are listed in Tab. 1, where this work is compared to other relevant state-of-the-art high-speed receivers. Prior demonstrations of monolithic receivers [8, 10, 14] are limited to gross line rates of 160 Gb/s. The presented receiver also exhibits the highest power efficiency.

The presented results establish monolithic SiGe BiCMOS EPIC technology as a viable platform for next-generation 400-Gb/s-per-lane optical receivers.

Tab. 1: Comparison with state-of-the-art optical receivers.

	[8]	[10]	[14]	[15]	This work
Monolithic	Yes	Yes	Yes	No	Yes
Symbol Rate [GBd]	56	50	160	160	224
Z_T [dB Ω]	PAM4	NRZ	NRZ	PAM4	PAM4
OE BW [GHz]	68.5 ^a	48.9 ^a	51	77	61.8 ^a
GDV [ps]	26 ^a	40.3 ^a	>110	64	>100
Noise [pA/ $\sqrt{\text{Hz}}$]	-	17 ^a	-	20.3	11.2 ^a
Power [pJ/bit]	17.7 ^a	17.5 ^a	-	18.8	22.7 ^a
	1.0	1.08	1.1	1.5	0.4

^aBased on simulation.

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